

Treasury Products — Gold Reference

Complete reference for every product in the market-risk-engine drills. Ordered **simple** → **complex**. Each entry: **(1) what it is** → **(2) payoff** → **(3) valuation formula** → **(4) toy example** → **(5) intuition** → **(6) closer (par strike + risk factors)**.

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Universal symbols

N	notional (your stake), in domestic currency unless flagged "foreign"
T	maturity in years
τ	year fraction of one accrual period (1.0 annual, 0.25 quarterly, etc.)
K	STRIKE / contracted rate – locked at trade time, never changes
DF(t)	discount factor today for cash arriving at year t $(= \exp(-(z(t) \cdot t))$
DF(t,s)	spread-adjusted discount factor $(= \exp(-(z(t)+s) \cdot t))$
S, fx	spot FX rate today (domestic-per-foreign)
b	breakeven inflation rate today (market's implied)
σ	implied volatility
N(·)	standard-normal CDF (for Black-Scholes-style options)
A(T)	annuity factor over a coupon strip $(= \sum \tau \cdot DF(t_i))$

Master skeleton for linear products: $V = (\text{market_implied} - \text{contract}) \cdot \text{scale} \cdot \text{discount}$.

Reference curve used in every worked example

z(1y)	= 4.50%	→	DF(1)	= 0.95600	
z(2y)	= 4.40%	→	DF(2)	= 0.91576	
z(3y)	= 4.30%	→	DF(3)	= 0.87897	
z(4y)	= 4.25%	→	DF(4)	= 0.84366	(interpolated)
z(5y)	= 4.20%	→	DF(5)	= 0.81058	
z(10y)	= 4.10%	→	DF(10)	= 0.66365	
$\sum DF(1..5)$					= 4.40498 ← annuity factor for \$10–\$13

DFs shown to 5 places; numerical outputs computed from full-precision values.

Quick-jump matrix (find anything in 5 seconds)

Product	\$	Pattern	Primary risk factor
FX Spot	1	none (direct)	FX delta
Repo / Reverse Repo	2	discount × notional	rate
Sec Lending	3	B (basis × annuity)	rate (small)
Amortising Loan	4	C (cashflow strip)	rate + credit spread
FX Forward	5	A (linear-in-strike)	FX delta + rate
FRA	6	A	rate
Vanilla Bond	7	C	rate + credit spread
Foreign Bond	8	C + FX	rate + credit + FX
Linker	9	C + inflation uplift	rate + breakeven
IRS	10	A (or C × 2 legs)	rate
ZCIS	11	A	rate + breakeven
Tenor Basis Swap	12	B	basis spread + rate
XCCY Basis Swap	13	B + FX	basis + FX + rate
TRS	14	reference-asset PV – notional	inherits from reference
CDS	15	A (with hazard rate substitution)	credit spread
FX Option	16	Black on forward	FX + vol + rate
Swaption	17	Black on swap rate	rate + vol
Annuity Liability	18	C (NEGATIVE sign)	rate + breakeven

Product	§	Pattern	Primary risk factor
Duration / DV01 / Convexity	19	metric, not product	derived from rate sensitivity
Extended Universe (T-bill, FRN, OIS, CMS, MBS, ...)	E1-E15	same patterns	varies

Part 0 — The Master Recipe (read this FIRST)

Every product in this document is a special case of **one identity**. Learn this and you can derive any valuation from scratch — no memorisation.

The single identity

$$V_0 = \sum_{t=t_1}^T E^Q[CF_t] \cdot DF(t)$$

(no-arbitrage / risk-neutral price)

"expected cashflow" "today's price of \$1 at t"
under the right risk-neutral measure (curve-derived)

That's it. Every PV formula in this document is $\sum (\text{expected CF}) \cdot (\text{discount})$. The trick is knowing **which substitution to apply** to each cashflow.

The 4-step recipe

- STEP 1. Write down every cashflow: when, in what currency, contingent on what?
- STEP 2. Replace each random cashflow with its expected value under the risk-neutral measure — see substitution table below.
- STEP 3. Discount each expected cashflow to today: multiply by $DF(t)$.
For credit-risky issuers, use the SPREAD-ADJUSTED $DF(t, s)$.
- STEP 4. Sum. $V = \sum$.

That's the whole skill. Steps 1, 3, 4 are bookkeeping. **Step 2 — the substitution — is the only piece that requires insight.**

The substitution table — what replaces each random CF

CF depends on...	Replace with	Why it works
Nothing (deterministic — e.g. bond coupon)	the cashflow itself	trivially deterministic
A future floating rate L_i (FRA, IRS float leg, FRN)	the curve's implied forward fwd_i	L_i is a martingale under the T_{i+1} - forward measure
An inflation-index ratio at T (ZCIS, Linker)	$(1 + b)^T$ where b is breakeven	breakeven IS the market's implied inflation
A spot FX rate at T (FX forward)	$S \cdot DF_{\text{for}}(T)/DF_{\text{dom}}(T)$ (the forward FX)	interest-rate parity (no-arb between currencies)
An option payoff $\max(S_T - K, 0)$ (caplet, FX option, swaption)	$DF(T) \cdot [F \cdot N(d1) - K \cdot N(d2)]$ — Black formula on the forward $\times$ discount	Black model: forward is the martingale, vol is known. The $DF(T)$ is the standard prefactor — already-discounted form.
A survival event up to t (CDS premium leg)	$Q(t) = \exp(-\lambda \cdot t)$ (survival probability)	hazard-rate model gives Q
A default event in $(t_{i-1}, t_i]$ (CDS protection leg)	$Q(t_{i-1}) - Q(t_i)$	survival difference
A SWAP RATE at t (CMS)	forward swap rate + convexity adjustment	the swap rate is NOT a martingale $\rightarrow$ needs a correction

The first 7 rows cover ~95% of treasury products. The 8th (CMS / quanto / non-trivial measure changes) is where the math gets harder and you need explicit measure-change machinery.

Three worked walk-throughs of the recipe

Walk-through 1 — FRA (replace floating rate with curve forward)

```
STEP 1. Cashflow:  pay  $N \cdot K \cdot \tau$           at  $T_2$           (you contracted)
                   receive  $N \cdot L \cdot \tau$        at  $T_2$           (L observed at  $T_1$ , random)
                   Net:  $N \cdot (L - K) \cdot \tau$       at  $T_2$ 

STEP 2. L is a floating rate – replace with the curve's implied forward:
        fwd =  $(DF(T_1)/DF(T_2) - 1) / \tau$ 

STEP 3. Discount the expected net to today:  $\cdot DF(T_2)$ 

STEP 4. Sum (single CF):
        V =  $N \cdot (\text{fwd} - K) \cdot \tau \cdot DF(T_2)$ 
```

That IS the FRA formula in §6. You didn't memorise it — you derived it.

Walk-through 2 — ZC Inflation Swap (replace realised inflation with breakeven)

```
STEP 1. Cashflow:  receive  $N \cdot (1 + \pi_{\text{real}})^T$  at T (random)
                   pay       $N \cdot (1 + K)^T$       at T (fixed)
                   Net:     $N \cdot [(1 + \pi_{\text{real}})^T - (1 + K)^T]$ 

STEP 2. Replace  $(1 + \pi_{\text{real}})^T$  with  $(1 + b)^T$  where b = market breakeven:
         $E^Q[(1 + \pi_{\text{real}})^T] = (1 + b)^T$ 

STEP 3. Discount to today:  $\cdot DF(T)$ 

STEP 4. Sum:
        V =  $N \cdot DF(T) \cdot [(1 + b)^T - (1 + K)^T]$ 
```

That's §11. Same recipe, different substitution.

Walk-through 3 — CDS (replace survival/default events with hazard probabilities)

```
STEP 1. Premium leg cashflows: pay  $N \cdot K \cdot \tau$  at each  $t_i$  IF still alive at  $t_i$ 
Protection leg:                receive  $N \cdot (1 - R)$  at default time  $\tau_{\text{default}}$ 

STEP 2. Replace "alive at  $t_i$ " with  $Q(t_i) = \exp(-\lambda t_i)$ 
        Replace "default in  $(t_{i-1}, t_i]$ " with  $Q(t_{i-1}) - Q(t_i)$ 

STEP 3. Discount each leg's expected CF to today:  $\cdot DF(t)$ 

STEP 4. Sum (premium leg – protection leg, from buyer's perspective):
        V_buyer =  $N \cdot [\sum (Q(t_{i-1}) - Q(t_i)) \cdot (1 - R) \cdot DF(t_i) - \sum K \cdot \tau \cdot Q(t_i) \cdot DF(t_i)]$ 
                                                         ← protection PV
                                                         ← premium PV
```

That's §15. Three recipes, three products — all derived top-down from the same identity.

The three "magic" substitutions you must internalise

These are the no-arbitrage SHORTCUTS that make Step 2 quick. Memorise these three:

1. FORWARD-RATE replacement:

$$E^Q[L(T)] = (DF(T_1) / DF(T_2) - 1) / \tau \quad \leftarrow \text{curve-implied forward}$$
 Use for: FRA, IRS float leg, FRN, caplet underlying.
2. RISK-NEUTRAL FORWARD replacement:

$$E^Q[S(T)] = S_0 \cdot DF_{\text{for}}(T) / DF_{\text{dom}}(T) \quad \leftarrow \text{FX forward}$$

$$E^Q[S(T)] = S_0 / DF(T) \quad \leftarrow \text{stock forward (with no divs)}$$
 Use for: FX forward, equity forward, option underlyings.
3. SURVIVAL-PROBABILITY replacement:

$$E^Q[1_{\{\text{still alive at } t\}}] = Q(t) = \exp(-\lambda t) \quad \leftarrow \text{hazard-rate model}$$
 Use for: CDS, credit-risky bonds, CVA on any product.

Plus the trivial fourth: **deterministic CF → itself** (bond coupons, principal, fixed-leg payments).

When the recipe needs adjustment (advanced — flag these and look up)

Most products in this doc work with steps 1-4 unchanged. A few require **convexity / quanto / measure-change corrections** the recipe doesn't capture:

Situation	Why the simple recipe fails	What to add
CMS (swap rate paid in cash)	Swap rate isn't a martingale under the payment measure	Convexity adjustment (Hagan replication via swaptions)
In-arrears FRA (rate paid AT fixing, not later)	Compounding date ≠ payment date → bias	Timing/convexity adjustment
Quanto option (foreign asset paying in domestic)	Foreign asset isn't a martingale in domestic measure	Quanto adjustment via correlation × vol × vol
YoY inflation (vs ZCIS)	Year-by-year payments under nominal measure	Inflation/rate correlation adjustment
Bermudan / American optionality	Holder optimally exercises → no analytical closed form	LSMC, lattice, PDE
Defaultable counterparty (any product)	Subject to your COUNTERPARTY'S default	CVA / DVA — bilateral xVA

Practical rule: if step 2 requires you to compute the expectation of a random variable under a measure where it ISN'T naturally a martingale, you need a convexity adjustment. The 99% case (FRA, IRS, ZCIS, vanilla bond, vanilla CDS) doesn't.

The "no-arbitrage replication" mental model (alternative derivation)

Equivalent way to derive any of these — DON'T use risk-neutral expectations, build the product from STATIC INSTRUMENTS instead.

Want to value: contract paying $f(\text{market})$ at T

Find a self-financing portfolio of TRADED instruments that pays the same $f(\text{market})$ at T .

No-arbitrage: contract value today = cost of the replicating portfolio today.

Example for FX Forward (parallel to walk-through 2): - Want: at T , pay K_{dom} and receive 1 foreign. - Replicate today: - Sell $DF_{\text{for}}(T)$ foreign bonds → receive $S \cdot DF_{\text{for}}(T)$ in domestic today; owe 1 foreign at T ✓ - Buy $K \cdot DF_{\text{dom}}(T)$ domestic bonds → pay $K \cdot DF_{\text{dom}}(T)$ today; receive K at T ✓ - Net cost today = $K \cdot DF_{\text{dom}}(T) - S \cdot DF_{\text{for}}(T)$. - For a 0-cost forward (par strike): set net = 0 → $K_{\text{par}} = S \cdot DF_{\text{for}}(T) / DF_{\text{dom}}(T)$. ✓

This is the "deposit-replication" argument from 03_curve_building.md. Both methods give the same answer; pick whichever is faster on a given problem.

How to use this when DERIVING a new product

1. Read the contract: list every cashflow, who pays whom, when, contingent on what.
2. For each cashflow, classify the underlying (deterministic / floating rate / FX / inflation / survival / option).
3. Apply the matching substitution from the table.
4. Multiply each expected CF by its $DF(t)$.
5. Sum.
6. Sanity check: does it reduce to a known case at boundary values?
(e.g. set $b = K \rightarrow$ ZCIS should give 0; set $\sigma \rightarrow 0 \rightarrow$ option $\rightarrow$ intrinsic value;
set $\lambda \rightarrow 0 \rightarrow$ CDS protection $\rightarrow 0$.)

Any cheatsheet entry below is just the result of running this 6-step process on a particular contract. You can re-derive any of them from a blank sheet.

The Portfolio Recipe (companion to the pricing recipe)

The Master Recipe values ONE contract. The Portfolio Recipe sizes a HEDGE BOOK across many contracts. Same 4-step structure, different verbs:

- STEP 1. EXPOSURE VECTOR.
For each portfolio position, compute its sensitivities – bucket by:
{DV01 per tenor (KRD: 2y, 5y, 10y, 30y), IE01, CS01, FX delta, vega per tenor}.
- STEP 2. AGGREGATE TO PORTFOLIO LEVEL.
Sum signed exposures across all instruments (assets POSITIVE, liabilities NEGATIVE) $\rightarrow$ net exposure vector for the book.
- STEP 3. CHOOSE A HEDGING SET.
Pick a small basket of liquid instruments (IRS strip at 2y/5y/10y/30y, Linker strip, nominal bonds, OIS, ZCIS) – one per factor you want to neutralise.
- STEP 4. SOLVE FOR HEDGE NOTIONALS.
Linear system: $HEDGE_NOTIONALS \cdot HEDGE_EXPOSURES = -NET_EXPOSURE$.
Residual: report any unhedged buckets (typically long-end KRD, AA-vs-swap basis, prepay risk) as the "stuff you can't hedge".

Sanity check: after applying the hedge, re-run a stress scenario (parallel ± 100 bp, steepener, +50 bp inflation, +50 bp credit) on the COMBINED book. Residual P&L should be $\ll$ unhedged. Anything that's not is your remaining basis risk — surface it explicitly.

This is the meta-skill for ALM, LDI, treasury risk: think in vectors of factors, hedge factor-by-factor, residualise what's left.

Part I — Cash & Money Market

1. FX Spot

What: own foreign currency. No maturity, no interest, no contract — you just hold the asset.

Payoff today: N_{for} units of foreign currency, worth $N_{\text{for}} \cdot S \cdot \text{fx_factor}$ in domestic.

Value today (domestic currency):

$$V = N_{\text{for}} \cdot S \cdot \text{fx_factor}$$

foreign today's FX rate
amount ($\times 1 + \text{fx_shock}$)

Toy example (FXSpot(80.0, 1.25)):

Input	Value	Meaning
N_for	80	own 80 units of foreign currency
S	1.25	spot = 1.25 (domestic per foreign)
fx_factor	1.0	no FX shock applied

$$V = 80 \cdot 1.25 \cdot 1.0 = 100.00$$

Intuition: pure FX exposure — every 1% FX move changes value by 1%. No rate sensitivity (no DF), no time (no τ or T).

Risk factors: FX (delta only). FX delta = $N_{for} \cdot S$.

2. Repo / Reverse Repo

What: secured short-term borrowing (repo) or lending (reverse repo). You sell a security today for cash, agree to buy it back at a higher price at maturity. The price difference IS the interest.

Payoff at T (reverse repo = lender): receive $N \cdot (1 + r_{\text{repo}} \cdot T)$.

Value today (PV of the maturity cashflow minus the cash you put up):

$$V = N \cdot (1 + r_{\text{repo}} \cdot T) \cdot DF(T) - N$$

PV of repaid cash + interest cash you advanced
(cost basis)

For the borrower (repo): the value is the negative — you owe that PV.

Toy example (Repo(100.0, 0.043, 0.5, is reverse=True)):

Input	Value	Meaning
N	100	\$100 of cash advanced
r_repo	0.043	repo rate 4.3% (simple interest, ACT/360-ish)
T	0.5	6-month repo
DF(0.5)	≈ 0.978	half-year discount factor from the curve

$$\begin{aligned}
 \text{maturity cash} &= 100 \cdot (1 + 0.043 \cdot 0.5) = 102.15 \\
 \text{PV} &= 102.15 \cdot 0.978 = 99.88 \\
 V &= 99.88 - 100 = -0.12
 \end{aligned}$$

Intuition: at trade time, repo rate $\approx$ short funding rate $\rightarrow V \approx 0$. The small residual reflects that the contracted repo rate differs slightly from the curve's implied funding rate. **If the repo rate > funding rate** $\rightarrow$ reverse repo is a positive NPV trade (you lent above market).

Risk factors: rate (very small — short tenor $\rightarrow$ tiny DV01).

3. Securities Lending

What: lend out securities you own; receive a fee every period.

Payoff at each t_i : receive $(\text{fee_bp}/10000) \cdot N \cdot \tau$ paid at the end of each period.

Value today:

$$V = \underbrace{(\text{fee_bp} / 10000)}_{\text{fee in decimal}} \cdot \underbrace{N}_{\text{notional}} \cdot \underbrace{\sum \tau \cdot \text{DF}(t_i)}_{\substack{\text{annuity factor} \\ (\text{sum of DFs})}}$$

Toy example (`SecLending(100.0, 20.0, (1,2,3), 1.0)`):

Input	Value	Meaning
N	100	\$100 of securities lent
fee_bp	20.0	20 bp annual fee
times	(1,2,3)	fees paid at end of years 1, 2, 3
τ	1.0	annual accrual
$\sum \tau \cdot \text{DF}$	$0.95600 + 0.91576 + 0.87897 = 2.75073$	3-year annuity factor

$$V = 0.002 \cdot 100 \cdot 2.75073 = 0.5501$$

Intuition: positive PV equal to the PV of three years of 20 bp fees. Pure "basis $\times$ annuity" pattern — same skeleton as the basis swaps.

Risk factors: rate (via annuity DFs — tiny DV01).

4. Amortising Loan

What: you lent money; the borrower repays equal principal each period plus interest on the remaining balance. (Differs from a mortgage's level-payment annuity — this is straight-line principal amortisation.)

Payoff each period (year i of n): receive $\text{amort} + c \cdot \text{outstanding}_{\{i-1\}}$ where $\text{amort} = \text{principal}/n$.

Value today (each cashflow discounted at a SPREAD-ADJUSTED rate — borrower's credit risk):

$$V = \sum_{i=1}^n \text{cf}_i \cdot \text{DF}(i, \text{spread_bp})$$

PV of declining-balance interest +
constant principal amortisation,
discounted at credit-adjusted curve

Toy example (`Loan(100.0, 0.05, 5, 200.0)`):

Input	Value	Meaning
principal	100	\$100 lent
coupon	0.05	5% on remaining balance each year
n	5	5 annual amortisation payments
spread_bp	200	borrower spread 200 bp (credit risk)
amort	20	\$20 principal repayment each year

```

year 1: cf = 20 + 0.05·100 = 25      (outstanding now 80)
year 2: cf = 20 + 0.05·80  = 24      (outstanding now 60)
year 3: cf = 20 + 0.05·60  = 23      (outstanding now 40)
year 4: cf = 20 + 0.05·40  = 22      (outstanding now 20)
year 5: cf = 20 + 0.05·20  = 21      (outstanding now 0)

```

```

V = Σ cf_i · DF(i, spread+200bp) ≈ 96.12

```

Intuition: $PV < \$100$ principal because the borrower's spread (200 bp) is **higher than the loan coupon** (5% vs ~4.2% risk-free + 2% spread = 6.2% required) — so this loan is "under-coupon" for its credit risk, and trades below par. **If coupon > risk-free + spread** → loan trades above par.

Risk factors: rate (DV01), credit spread (CS01 — bump spread_bp +1).

Part II — Forwards

5. FX Forward

What: agree TODAY to swap currencies at a pre-agreed rate K on date T .

Payoff at T : pay $N_{\text{for}} \cdot K$ domestic, receive N_{for} foreign.

Value today (mark-to-market):

$$V = N_{\text{for}} \cdot (\underset{\substack{\text{size} \\ \text{(per unit of foreign)}}}{S \cdot \text{fx_factor}} - K) \cdot \underset{\substack{\text{today's spot vs contract} \\ \text{PV}}}{DF(T)}$$

Toy example ($\text{FXForward}(100.0, 1.25, 1.28, 1.0)$):

Input	Value	Meaning
N_{for}	100	will buy 100 foreign at maturity
S	1.25	today's spot
K	1.28	contracted rate (locked at trade time)
T	1.0	1 year
$DF(1)$	0.95600	one-year discount factor

$$V = 100 \cdot (1.25 - 1.28) \cdot 0.95600 = -2.87$$

Intuition: you agreed to BUY foreign at 1.28 but it's worth 1.25 today $\rightarrow$ ~3 cents loss per unit, PV'd back.
Output = what you'd pay today to cancel.

Par strike: $K_{\text{par}} = S \cdot \text{fx_factor}$ (simplified drill form) | $K_{\text{par}} = S \cdot DF_{\text{for}}(T)/DF_{\text{dom}}(T)$ (full IRP).

Risk factors: FX delta, rate via $DF(T)$.

6. FRA — Forward Rate Agreement

What: agree TODAY on the interest rate K to apply to a future deposit period $[T_1, T_2]$.

Payoff at T_2 (receive-floating, pay-fixed = long FRA): $N \cdot (L - K) \cdot \tau$ where L is the fixing at T_1 .

Value today (replace unknown L with the curve's implied forward):

$$\text{fwd} = \left(\frac{DF(T_1)}{DF(T_2)} - 1 \right) / \tau \quad \leftarrow \text{simple-comp forward}$$

$$V = \underbrace{N}_{\text{size}} \cdot \underbrace{(\text{fwd} - K)}_{\substack{\text{market vs} \\ \text{contract}}} \cdot \underbrace{\tau}_{\substack{\text{accrual} \\ \text{factor}}} \cdot \underbrace{DF(T_2)}_{\text{PV}}$$

Toy example ($\text{FRA}(100.0, 1.0, 2.0, 1.0, 0.043)$):

Input	Value	Meaning
N	100	notional
T_1, T_2	1.0, 2.0	forward window: year 1 to year 2

Input	Value	Meaning
τ	1.0	annual
K	0.043	locked rate 4.30%
fwd	$(0.95600/0.91576 - 1)/1.0 = 0.04394$	market-implied forward

$$V = 100 \cdot (0.04394 - 0.043) \cdot 1.0 \cdot 0.91576 = 0.0859$$

Intuition: market is pricing 4.39% but you locked in 4.30% — you're 9 bp better off, scaled by notional $\times \tau$, discounted.

Par strike: $K_{\text{par}} = \text{fwd}$. **Risk factors:** rate only.

MTM vs settlement: the formula above is the curve-based MTM. In real settlement the cash changes hands at T_1 discounted at the observed fixing L : $\text{payment} = N \cdot (L - K) \cdot \tau / (1 + L \cdot \tau)$. The two agree in expectation under the T_2 -forward measure; the MTM form is simpler because it only needs the curve.

Part III — Bonds

7. Vanilla Bond

What: a coupon-paying bond — issuer promises a stream of cashflows on fixed dates.

Payoff each t_i : receive cf_i (coupons on most dates, coupon + principal at maturity).

Value today (each cashflow discounted at spread-adjusted curve = issuer's credit risk):

$$V = \sum_{\substack{\uparrow \\ \text{sum over all cashflow dates}}} cf_i \cdot DF(t_i, \text{spread_bp}) \quad DF(t, s) = \exp(-(z(t) + s/1e4) \cdot t)$$

Toy example (`Bond((1,2,3,4,5), (4,4,4,4,104), 150.0)`):

Input	Value	Meaning
times	(1,2,3,4,5)	annual coupon dates
cashflows	(4, 4, 4, 4, 104)	\$4 coupons + \$100 principal at year 5
spread_bp	150	issuer spread = 150 bp over the risk-free curve

```
DF(1, +150bp) = exp(-(0.045 + 0.015)·1) = 0.9418    PV_1 = 4 · 0.9418 = 3.767
DF(2, +150bp) = exp(-(0.044 + 0.015)·2) = 0.8884    PV_2 = 4 · 0.8884 = 3.554
DF(3, +150bp) = exp(-(0.043 + 0.015)·3) = 0.8404    PV_3 = 4 · 0.8404 = 3.361
DF(4, +150bp) = exp(-(0.0425+0.015)·4) = 0.7945     PV_4 = 4 · 0.7945 = 3.178
DF(5, +150bp) = exp(-(0.042 + 0.015)·5) = 0.7515     PV_5 = 104 · 0.7515 = 78.211
```

$V \approx 92.07$

Intuition: standard PV of fixed cashflows. The spread (150 bp) is the issuer's credit premium — wider spread → lower DFs → lower price. Below par because the coupon (4%) is lower than the issuer's required yield (~5.7% = risk-free + 150 bp).

Par coupon (the coupon that prices the bond at 100): solve $\sum c \cdot DF(t,s) + 100 \cdot DF(T,s) = 100 \rightarrow c = (1 - DF(T,s)) / \sum DF(t,s)$. Same identity as the IRS par swap rate (\$10).

Risk factors: rate (DV01), credit spread (CS01).

8. Foreign-Currency Bond

What: a bond denominated in foreign currency. PV is the local-currency PV translated to domestic at spot.

Value today (domestic currency):

$$V = \left(\sum cf_i \cdot DF(t_i, \text{spread_bp}) \right) \cdot S \cdot \text{fx_factor}$$

local-currency PV (like \$7) FX conversion

Toy example (`ForeignBond((1,2,3,4,5), (3,3,3,3,103), 100.0, 1.25)`):

Input	Value	Meaning
times, cashflows	(1..5), (3,3,3,3,103)	3% coupons + principal in foreign
spread_bp	100	issuer spread 100 bp
S, fx_factor	1.25, 1.0	spot, no FX shock

Local PV $\approx$ 89.94 (compute like \$7 but with spread=100bp)
 $V = 89.94 \cdot 1.25 \cdot 1.0 = 112.43$

Intuition: same as Vanilla Bond but multiplied by current FX rate. Risk = sum of foreign-curve risk AND FX delta.

Risk factors: rate (foreign curve via DFs), credit spread, FX (delta = local PV).

9. Linker — Inflation-Linked Bond

What: bond whose every cashflow is uplifted by accumulated inflation between issue and payment.

Payoff at each t_i : real coupon c_t $\times$ index ratio $(1 + \pi_{\text{realised}})^t$.

Value today (replace realised inflation with market breakeven b):

$$V = \sum \text{real cashflow}_i \cdot (1 + b)^{t_i} \cdot \text{discount}(t_i) \quad \text{(nominal curve)}$$

Toy example (Linker((1,2,3,4,5), (1.5,1.5,1.5,1.5,101.5), 0.03)):

Input	Value	Meaning
times	(1..5)	annual
real_cfs	(1.5, 1.5, 1.5, 1.5, 101.5)	1.5% real coupons + 100 principal at year 5
b	0.03	breakeven inflation 3%

```

y1: 1.5 · 1.03^1 · 0.95600 = 1.477
y2: 1.5 · 1.03^2 · 0.91576 = 1.457
y3: 1.5 · 1.03^3 · 0.87897 = 1.441
y4: 1.5 · 1.03^4 · 0.84366 = 1.424
y5: 101.5 · 1.03^5 · 0.81058 = 95.379
      V ≈ 101.178

```

Intuition: standard bond PV but cashflows are uplifted by $(1+b)^t$ — higher breakeven → higher payments → higher PV. Linker is **long inflation**.

Risk factors: rate (DV01 via DFs), inflation breakeven (IE01).

Indexation lag: in production, the inflation index used at payment date is typically the CPI level from 3M earlier (UK index-linked gilts) or 8M earlier (US TIPS) — adds a small basis between true real cashflows and contractual ones. Drill formula ignores this.

Part IV — Swaps

10. Interest Rate Swap (IRS)

What: agree TODAY to exchange a fixed-rate stream (c on N per period) for a floating-rate stream (L_i on N per period) on the same dates.

Payoff each t_i (receive-fixed): receive $N \cdot c \cdot \tau$, pay $N \cdot L_i \cdot \tau$.

Value today (receive-fixed perspective; uses telescoping identity for the floating leg):

$$V = N \cdot \left[\underbrace{c \cdot \tau \cdot \sum DF(t_i)}_{\text{fixed leg PV}} - \underbrace{(1 - DF(T))}_{\text{floating leg PV (single-curve telescope)}} \right]$$

($N \cdot c \cdot A(T)$)

For pay-fixed, flip the sign.

Toy example (`InterestRateSwap((1,2,3,4,5), 1.0, 0.042, 100.0, receive_fixed=True)`):

Input	Value	Meaning
N	100	notional
times, τ	(1..5), 1.0	5 annual payments
c	0.042	contracted fixed rate 4.20%
$\sum DF(1..5)$	4.40498	annuity factor
$DF(5)$	0.81058	

$$\begin{aligned} \text{fixed leg PV} &= 100 \cdot 0.042 \cdot 4.40498 = 18.501 \\ \text{float leg PV} &= 100 \cdot (1 - 0.81058) = 18.942 \\ V &= 18.501 - 18.942 = -0.441 \end{aligned}$$

Intuition: you contracted to receive 4.20% but the curve's par rate today is higher (~4.30%), so you're under-paid by ~10 bp on \$100 over 5 years — small negative PV.

Par swap rate: $c_{\text{par}} = (1 - DF(T)) / \sum \tau \cdot DF(t_i)$ — the rate that makes the swap worth zero today.

Risk factors: rate only (huge DV01 vs other products — IRS is THE rates instrument).

11. Zero-Coupon Inflation Swap (ZCIS)

What: agree TODAY to exchange a single payment of $N \cdot (1 + \pi_{\text{realised}})^T$ (floating, inflation) for $N \cdot (1 + K)^T$ (fixed) at maturity T .

Payoff at T (receiver of inflation): $N \cdot [(1 + \pi_{\text{realised}})^T - (1 + K)^T]$.

Value today (replace realised inflation with breakeven):

$$V = \underbrace{N}_{\text{size}} \cdot \underbrace{DF(T)}_{\text{PV}} \cdot \underbrace{[(1 + b)^T - (1 + K)^T]}_{\text{market-compound vs contract-compound}}$$

Toy example (`InflationSwap(100.0, 5.0, 0.03, 0.028)`):

Input	Value	Meaning
N	100	
T	5.0	matures in 5y

Input	Value	Meaning
b	0.03	breakeven 3%
K	0.028	locked 2.8%

$$\begin{aligned}
 (1+b)^T &= 1.03^5 = 1.15927 \\
 (1+K)^T &= 1.028^5 = 1.14803 \\
 \text{gap} &= 0.01124
 \end{aligned}$$

$$V = 100 \cdot 0.81058 \cdot 0.01124 \approx 0.909$$

Intuition: locked in 2.8% inflation, market is now pricing 3.0% — you gain ~20 bp compounded over 5y, discounted.

Par strike: $K_{\text{par}} = b$ (formula is symmetric — same $(1+\cdot)^T$ on both sides). **Risk factors:** rate (via $DF(T)$), inflation breakeven.

12. Tenor Basis Swap

What: exchange one floating-rate stream (e.g. 3M LIBOR) for another (e.g. 6M LIBOR) plus a spread. The "basis" is the small spread that compensates one side for the tenor mismatch.

Payoff per period (receiver of basis): $N \cdot \text{basis} \cdot \tau$ (the floating legs cancel up to the spread).

Value today:

$$V = b \cdot N \cdot \sum \tau \cdot DF(t_i) \quad \text{with } b = (\text{basis_bp} + \text{tenor_shock_bp}) / 1e4$$

basis in decimal

Toy example (`TenorBasisSwap(100.0, (1,2,3,4,5), 1.0, 15.0)`):

Input	Value	Meaning
N	100	
basis_bp	15	receive 15 bp on top of the equivalent leg
$\sum \tau \cdot DF$	4.40498	

$$\begin{aligned}
 b &= 0.0015 \\
 V &= 0.0015 \cdot 100 \cdot 4.40498 = 0.6607
 \end{aligned}$$

Intuition: pure stream of basis-spread payments → annuity × basis × notional.

Quick rule: every 1 bp of basis on \$N over the strip is worth $N \cdot A(T) \cdot 0.0001$ per bp. **Risk factors:** rate (annuity DFs), tenor-basis spread.

13. XCCY Basis Swap

What: cross-currency basis swap — exchange floating in one currency for floating in another. The basis is the spread that compensates for the currency mismatch (always-positive in practice, FX-hedged cost of foreign funding).

Payoff: same as tenor basis swap but the basis leg is in **foreign currency** → FX-sensitive.

Value today (domestic currency):

$$V = \underset{\text{foreign notional}}{N_{\text{for}}} \cdot \underset{\text{FX conversion}}{S} \cdot \underset{\text{FX conversion}}{fx_factor} \cdot b \cdot \sum \tau \cdot DF(t_i)$$

(Principal exchange omitted — included in real-world deals; see caveat at end of doc.)

Toy example (XCCYBasisSwap(100.0, 1.25, (1,2,3,4,5), 1.0, -20.0)):

Input	Value	Meaning
N_for	100	foreign notional
S	1.25	
basis_bp	-20	foreign side pays 20 bp (negative basis)
$\sum \tau \cdot DF$	4.40498	

$$\begin{aligned} b &= -0.002 \\ V &= 100 \cdot 1.25 \cdot (-0.002) \cdot 4.40498 = -1.1012 \end{aligned}$$

Intuition: same as tenor basis but in foreign currency. Negative basis = paying the spread → negative PV. Sensitive to FX too (the basis leg's PV depends on S).

Risk factors: rate (annuity DFs), XCCY-basis spread, FX delta.

14. Total Return Swap (TRS)

What: one party (TRS receiver) gets the total return of a reference asset (price changes + coupons); the other gets a funding rate on a notional. Essentially synthetic ownership.

Payoff: receiver of total return gets $PV(\text{reference}) - \text{funding_notional}$.

Value today:

$$V = \begin{array}{ll} PV(\text{reference asset}) & - \text{funding_notional} \\ \text{reprice using current} & \text{the financing} \\ \text{market state} & \text{leg's notional} \end{array}$$

Toy example (TotalReturnSwap(reference=Bond((1..5),(4,4,4,4,104),150.0), funding_notional=90.0)):

Input	Value	Meaning
reference	a Bond	the underlying being synthetically owned
funding_notional	90	financing leg notional (= what receiver "borrows")
PV(reference)	92.07	from §7 (Bond)

$$V = 92.07 - 90.0 = 2.07$$

Intuition: you're synthetically long the bond at 90, but it's worth 92 → you're up 2. As the bond rallies, you gain; as it sells off, you lose. **Same risk exposure as owning the bond**, but balance-sheet-light (no funding needed up front).

Risk factors: identical to the reference asset (rate, spread, etc.).

Part V – Credit

15. Credit Default Swap (CDS)

What: insurance against an issuer's default. Buyer pays a periodic premium K (the contractual spread); receives $(1 - R) \cdot N$ if default occurs.

Payoff: - buyer pays $N \cdot K \cdot \tau \cdot Q(t_i)$ each period until default or maturity (premium leg) - buyer receives $(1 - R) \cdot N$ at the default time (protection leg)

Value today (buyer perspective, hazard rate λ implied from current market spread):

$$\begin{aligned}\lambda &= \text{market_spread} / (1 - R) && \leftarrow \text{credit triangle} \\ \text{premium leg PV} &= \sum \tau \cdot DF(t_i) \cdot Q(t_i) && \text{where } Q(t) = \exp(-\lambda \cdot t) \\ \text{protection leg PV} &= \sum (Q(t_{i-1}) - Q(t_i)) \cdot DF(t_i) \\ V_{\text{buyer}} &= N \cdot [\text{protection} \cdot (1 - R) - \text{premium} \cdot K]\end{aligned}$$

Toy example (CreditDefaultSwap((1..5), 1.0, contract=100bp, market=120bp, N=100)):

Input	Value	Meaning
contract_K	100 bp	you contracted to pay 100 bp/yr
market spread	120 bp	market is now pricing 120 bp
R	0.4	assumed 40% recovery
N	100	

$$\begin{aligned}\lambda &= 0.012 / 0.6 = 0.02 \\ \text{Premium} &= \sum \tau \cdot DF \cdot Q \approx 4.07 \\ \text{Protect} &= \sum (Q_{i-1} - Q) \cdot DF \approx 0.083 \\ V_{\text{buyer}} &= 100 \cdot [0.083 \cdot 0.6 - 4.07 \cdot 0.01] \\ &= 100 \cdot [0.050 - 0.0407] \\ &\approx 0.88\end{aligned}$$

Intuition: you locked in 100 bp protection; the market now charges 120 bp → your contract is "cheap protection" by 20 bp/yr → positive PV. **Spread widens → buyer wins** (protection becomes more valuable).

Par spread: $K_{\text{par}} = \text{market_spread}$ (set $V=0 \rightarrow$ contract equals market). **Risk factors:** credit spread (CS01 dominates), rate (small DV01).

Part VI — Options

16. FX Option

What: right (not obligation) to exchange currencies at strike K on expiry T . Call = right to BUY foreign; put = right to SELL foreign.

Payoff at T (call): $\max(S_T - K, 0) \cdot N$.

Value today (Black formula — uses the forward F directly):

$$\begin{aligned} F &= S \cdot \text{fx_factor} / \text{DF}(T) && \leftarrow \text{forward FX rate} \\ v &= \sigma + \text{vol_shock} && \leftarrow \text{stressed vol} \\ d1 &= [\ln(F/K) + \frac{1}{2} \cdot v^2 \cdot T] / (v \cdot \sqrt{T}) \\ d2 &= d1 - v \cdot \sqrt{T} \\ V_{\text{call}} &= N \cdot \text{DF}(T) \cdot [F \cdot N(d1) - K \cdot N(d2)] \\ V_{\text{put}} &= N \cdot \text{DF}(T) \cdot [K \cdot N(-d2) - F \cdot N(-d1)] \end{aligned}$$

Toy example (FXOption(expiry=1.0, strike=1.30, vol=0.12, fx0=1.25, N=100, call=True)):

Input	Value	Meaning
S	1.25	today's spot
K	1.30	option strike
σ	12%	implied vol
T	1.0	1 year
DF(1)	0.95600	

$$\begin{aligned} F &= 1.25 / 0.95600 = 1.3075 \\ d1 &= [\ln(1.3075/1.30) + \frac{1}{2} \cdot 0.0144] / (0.12) \approx 0.108 \\ d2 &= d1 - 0.12 \approx -0.012 \\ N(d1) &\approx 0.543 \quad N(d2) \approx 0.495 \\ V_{\text{call}} &= 100 \cdot 0.95600 \cdot [1.3075 \cdot 0.543 - 1.30 \cdot 0.495] \\ &= 100 \cdot 0.95600 \cdot [0.710 - 0.644] \\ &\approx 6.33 \end{aligned}$$

Intuition: out-of-the-money call ($K > S$), but the forward is essentially ATM ($F = 1.3075$ vs $K = 1.30$) due to interest-rate differential, so the option has real time value. Output = today's premium.

Greeks: - **Delta** = $\text{DF}(T) \cdot N(d1)$ (call) $\rightarrow$ "shares of spot to hedge" - **Gamma** = $\text{DF}(T) \cdot \phi(d1) / (F \cdot v \cdot \sqrt{T} \cdot \text{DF}(T)) \rightarrow$ curvature - **Vega** = $N \cdot \text{DF}(T) \cdot F \cdot \phi(d1) \cdot \sqrt{T} \rightarrow$ vol sensitivity (per 1.0 vol point) - **Theta** = negative for long options $\rightarrow$ time decay

Risk factors: FX (delta), vol (vega), rate (rho).

17. Swaption

What: option to enter an interest rate swap at expiry T_{expiry} at strike rate K . Payer swaption = right to enter as fixed-PAYER; receiver swaption = right to enter as fixed-RECEIVER.

Value today (Black model on the forward swap rate):

```

A    =  $\sum \tau \cdot DF(t_i)$                                 ← swap annuity
S    =  $[ DF(T_{\text{expiry}}) - DF(T_{\text{end}}) ] / A$           ← forward swap rate
v    =  $\sigma + \text{vol\_shock}$                                 ← stressed vol on swap rate

d1   =  $[ \ln(S/K) + \frac{1}{2} \cdot v^2 \cdot T ] / ( v \cdot \sqrt{T} )$ 
d2   =  $d1 - v \cdot \sqrt{T}$ 

V_payer    =  $N \cdot A \cdot [ S \cdot N(d1) - K \cdot N(d2) ]$ 
V_receiver =  $N \cdot A \cdot [ K \cdot N(-d2) - S \cdot N(-d1) ]$ 

```

Toy example (Swaption(expiry=2.0, pay_times=(3,4,5), τ =1.0, K=0.042, σ =0.20, N=100, payer=True)):

Input	Value	Meaning
T_expiry	2.0	swaption exercise at year 2
pay_times	(3,4,5)	swap pays at years 3, 4, 5
K	0.042	option to enter at fixed rate 4.20%
σ	0.20	implied vol (lognormal Black)

```

A    = DF(3) + DF(4) + DF(5)          = 2.53321
S    = [DF(2) - DF(5)] / A             = [0.91576 - 0.81058] / 2.53321 = 0.04153
d1   = [ln(0.04153/0.042) + ½·0.04·2] / (0.20·√2) ≈ 0.101
d2   = d1 - 0.20·√2                    ≈ -0.182

V_payer = 100 · 2.53321 · [0.04153·0.540 - 0.042·0.428] ≈ 1.13

```

Intuition: option to PAY 4.20% for 3 years starting in 2 years. Forward swap rate today is 4.15% — so the option is slightly out-of-the-money for the payer, but still has time value due to vol.

Greeks: same families as FX option, but **vega** is the dominant exposure for at-the-money swaptions. **Risk factors:** rate (delta + rho via DFs), vol (vega).

Part VII — Liabilities

18. Annuity Liability

What: pension-style obligation — you OWE inflation-linked payments every year for `years` years.

Payoff each year (negative for you, the holder): $-\text{payment} \cdot (1 + \pi_{\text{realised}})^t$.

Value today (replace realised inflation with breakeven; **negative sign because liability**):

$$V = - \text{payment} \cdot \sum_{t=1}^{\text{years}} \underbrace{(1 + b)^t \cdot DF(t)}_{\text{inflation-uplifted annuity}}$$

Toy example (`AnnuityLiability(5.0, 10, 0.03)`):

Input	Value	Meaning
payment	5.0	\$5 per year real
years	10	
b	0.03	

```

y1: -5 · 1.03^1 · 0.95600 = -4.923
y2: -5 · 1.03^2 · 0.91576 = -4.857
y3: -5 · 1.03^3 · 0.87897 = -4.804
y4: -5 · 1.03^4 · 0.84366 = -4.747
y5: -5 · 1.03^5 · 0.81058 = -4.701
y6: -5 · 1.03^6 · 0.78198 = -4.671
y7: -5 · 1.03^7 · 0.75417 = -4.640
y8: -5 · 1.03^8 · 0.72714 = -4.611
y9: -5 · 1.03^9 · 0.70088 = -4.582
y10: -5 · 1.03^10 · 0.66365 = -4.464
      V ≈ -46.79

```

Intuition: 10 years of \$5 inflation-linked obligations. **Rates down → liability more negative, inflation up → liability more negative.** The negative sign matters: a portfolio of assets + liability is $V_{\text{assets}} + V_{\text{liab}}$ (liability is already negative).

Risk profile (THE pension ALM problem): - DV01 is **opposite-sign** of a long bond: rates down → liability's negative PV gets MORE negative → you OWE more - IE01 is **negative** (same sign as Linker on PV, but signed the other way): inflation up → liability more negative - Linker assets PARTIALLY hedge this — match IE01s to neutralise inflation

Hedge ratio mechanic: to neutralise IE01,

$$N_{\text{linker}} \cdot \text{IE01}_{\text{linker_per_unit_face}} = |\text{IE01}_{\text{liability}}|$$

where $\text{IE01}_{\text{liability}} = -\text{payment} \cdot \sum_t t \cdot (1+b)^{(t-1)} \cdot DF(t) \quad (\partial V / \partial b)$
 $\text{IE01}_{\text{linker}} = + \sum \text{cf_real_i} \cdot t_i \cdot (1+b)^{(t_i-1)} \cdot DF(t_i)$

Worked numerical hedge for our annuity (payment=5, 10y, b=3%) using a \$100-face 5y Linker (\$9, IE01 ≈ +0.05 per \$100 face — bump `b` by 1bp, reprice):

```

IE01_liab    ≈ -payment · Σ t · (1+b)^(t-1) · DF(t) · 0.0001
              ≈ -5 · 38.4 · 0.0001                      = -0.0192 per +1bp
IE01_linker  ≈ +0.045 per $100 face per +1bp
N_hedge      = |IE01_liab| / IE01_linker_per_$100 · 100    = $42.7

→ Hold $42.7 of 5y Linker face value for every $5/yr annuity. PARTIAL hedge
  (Linker's 5y duration < liability's 10y duration → uncovered long-end IE01).

```

Same logic for DV01 — match nominal duration with an IRS or nominal bond. But **parallel DV01 alone isn't enough**: a 10y liability hedged with a 5y IRS is parallel-neutral but **key-rate mismatched** at the long end. Real ALM uses **Key Rate Durations** (KRD at 2y/5y/10y/30y buckets) and matches each bucket separately. See `02_duration_convexity_krd.md` for the KRD machinery.

Sign convention note: under the convention $DV01 = -\partial V / \partial r \cdot 0.0001$ (positive for long bond), an annuity liability has **negative DV01** (rates $\uparrow \rightarrow V$ less negative $\rightarrow \partial V / \partial r > 0$, flipped sign $\rightarrow DV01 < 0$). The risk-factor table at the end of this doc uses the alternative $\partial V / \partial r$ convention which gives the liability a POSITIVE sign. Pick one across your stack.

Risk factors: rate (large DV01 — long duration on long-dated payments), breakeven inflation (large IE01).

Part VIII — Risk Metrics

19. Risk Metrics — Duration, Modified Duration, DV01, Convexity

These are sensitivities you compute ON TOP of any rate-bearing instrument (bonds, swaps, linkers, etc.) — they're not products themselves.

Macaulay Duration (D_{mac} , units = YEARS)

PV-weighted average time to cashflows. "How long until I get my money back on average."

$$D_{\text{mac}} = \frac{\sum t_i \cdot PV_i}{\sum PV_i} \quad \text{where } PV_i = cf_i \cdot DF(t_i)$$

For a 5y bond paying 4% coupons: $D_{\text{mac}} \approx 4.6$ years.

Modified Duration (D_{mod} , % price sensitivity)

The rate-sensitivity flavour — derivative of price w.r.t. yield, expressed in %:

$$D_{\text{mod}} = \frac{D_{\text{mac}}}{1 + y/m} \quad (\text{semi-annual bond: } m = 2)$$
$$\Delta P / P \approx -D_{\text{mod}} \cdot \Delta y \quad \leftarrow \text{per unit } \Delta y \text{ (e.g. } \Delta y = 0.01 = 1\%)$$

For the same bond: $D_{\text{mod}} \approx 4.45$. A 100 bp yield rise → price drops ~4.45%.

DV01 / PV01 (Dollar Value of 1 bp, units = \$)

Practical desk metric — dollar P&L per 1 bp rate move:

$$DV01 = D_{\text{mod}} \cdot P \cdot 0.0001$$

For \$100M of the bond: $DV01 \approx \$42,650$ per bp. Bond falls \$42k when rates rise 1 bp.

Convexity (C , second-order correction)

Curvature of the price-yield relationship — duration is only a tangent line; convexity is the next term in the Taylor expansion:

$$C = \frac{\sum t_i \cdot (t_i + 1/m) \cdot PV_i}{P \cdot (1 + y/m)^2}$$
$$\Delta P / P \approx -D_{\text{mod}} \cdot \Delta y + \frac{1}{2} \cdot C \cdot (\Delta y)^2$$

↑
always pushes IN YOUR FAVOR for long bonds
(curvature makes losses smaller, gains bigger)

Why convexity matters

For small moves duration alone is fine; for large moves you need convexity.

+100 bp move on a 30y bond:
duration alone says: -12% loss
actual: -11.5% loss (convexity saved you 0.5%)

Numerical (FD) versions — when there's no closed form

For exotic products (MC, lattice, basket), bump the rates and revalue:

```
DV01_numerical = V(rate + 1bp) - V(rate)          ← forward diff
                = [V(rate + 1bp) - V(rate - 1bp)] / 2 ← central diff (more accurate)

Convexity_FD    = [V(rate + h) - 2·V(rate) + V(rate - h)] / (h² · V(rate))
```

Greeks (for option-bearing products)

Generic non-rate sensitivities computed the same way (analytical for vanilla, FD for exotics):

Greek	What	Formula (Black on forward F)	Units
Δ delta	dV / dS	$DF(T) \cdot N(d1)$ (call)	\$/spot unit
Γ gamma	d^2V / dS^2	$DF(T) \cdot \phi(d1) / (F \cdot \sigma \cdot \sqrt{T})$	\$/spot ² (note: denominator uses F , not S)
ν vega	$dV / d\sigma$	$DF(T) \cdot F \cdot \phi(d1) \cdot \sqrt{T}$	\$/vol point (e.g. /1% vol)
Θ theta	dV / dt	(long expressions; numerical bump is easier)	\$/day
ρ rho	dV / dr	call-rho > 0	\$/rate unit

Convention: Gamma in the Black model uses the **forward F** in the denominator, not the spot S . The two differ by $F/S = 1/DF(T)$ for non-dividend assets — small for short tenors, material at 5y+. Many textbooks abuse notation by writing S where F belongs; verify which model you're using.

Part IX — Extended Treasury Universe (abbreviated entries)

Real treasury desks trade ~50+ products. The 18 above cover all major shapes; this section adds 15 more **variations on those shapes** — anything not here is a recombination of these primitives.

Format note: entries below are **abbreviated** (what / formula / use / risk factors). They share the same patterns as the worked products in Parts I-VII. For an example of the full 6-block template, see any of \$1-\$18. Each entry is prefixed E1, E2, ... (NOT continuing \$19) so it doesn't collide with the Risk Metrics section.

E1. T-Bill / Discount paper (short-term zero-coupon govt)

What: government short-term debt issued at a DISCOUNT (no coupons). Mature in 4w, 13w, 26w, 52w. Quoted on a discount-yield basis: $dy = (face - price) / face \cdot 360 / days$.

Value today = $face \cdot DF(T)$ (zero-coupon bond, no spread for risk-free)

Use: cash management, collateral, risk-free benchmark. **Risk factors:** rate only (small DV01).

E2. Commercial Paper (CP) / Banker's Acceptance

What: short-term corporate (CP) or bank (BA) zero-coupon debt. Same math as T-bill but with credit spread.

Value today = $face \cdot DF(T, spread_bp)$

Use: corporate funding (CP), trade finance (BA). **Risk factors:** rate, credit spread (CS01).

E3. Zero-Coupon Bond (long-dated)

What: bond with NO coupons, only the face at maturity T . Pure interest-rate play.

Value today = $face \cdot DF(T, spread_bp)$
Duration = T (Macaulay duration equals the maturity exactly – no PV-weighting needed)

Use: long-duration hedging (pension liabilities), TIPS strips. **Risk factors:** rate (very large DV01 because all weight at T), credit spread.

E4. Floating-Rate Note (FRN)

What: bond paying coupons indexed to a floating rate (LIBOR/SOFR + spread) every period.

At each t_i : pays $(L_i + spread) \cdot \tau \cdot face$
At maturity: pays $face$

Value today (just after a fixing) = $face \cdot spread \cdot \sum \tau \cdot DF(t_i) + face$
spread $\times$ annuity ← floating part is par

Caveat: the formula above assumes you're valuing AT a fixing date. Between fixings, the just-set coupon $L_{last} + spread$ is already known and accrues to the NEXT payment — the floater PV's to $(L_{last} + spread) \cdot \tau \cdot face \cdot DF(t_{next}) + face \cdot DF(t_{next})$ for the residual until the next reset, NOT exactly to par. The bias is small (a few bp) over short reset windows.

Use: low-duration credit exposure — duration $\approx$ time-to-next-fixing (often < 0.25 years). Same idea as IRS floating leg, but you hold the notional.

Risk factors: spread (dominant), tiny DV01 (just to next fixing).

E5. Callable Bond

What: bond + embedded short call option (issuer can buy back early at par). Issuer benefits if rates fall (refinance cheaper) — bondholder loses.

$$\text{Value} = \text{Value}(\text{straight bond}) - \text{Value}(\text{call option on the bond})$$

Negative convexity — bondholder bears asymmetric risk. **Use:** corporate issuers (callable munis, sub debt).

Risk factors: rate, credit, **vol** (the option's vega), call schedule.

E6. Convertible Bond

What: bond + embedded long call option on the issuer's equity (holder can convert into shares).

$$\text{Value} = \text{Value}(\text{straight bond}) + \text{Value}(\text{conversion option})$$

Hybrid risk: bond-like when far-from-conversion, equity-like when in-the-money. **Use:** tech / growth issuers.

Risk factors: rate, credit, equity, equity vol — multi-factor.

E7. Overnight Index Swap (OIS)

What: IRS where the floating leg is the COMPOUNDED overnight rate (Fed Funds, SOFR, €STR, SONIA) instead of LIBOR/term-rate. The benchmark "risk-free" swap.

Same formula as IRS (§10), but floating leg references the overnight curve.
Post-2008: OIS curve is the DISCOUNTING curve in multi-curve frameworks.

Use: hedge against central-bank rate moves, discount swap collateral, FX swap pricing. **Risk factors:** rate (OIS DV01).

E8. Constant-Maturity Swap (CMS)

What: IRS where the floating leg references a SWAP RATE of fixed tenor (e.g. 10y swap rate, observed every reset) instead of a short rate. Used to bet on curve shape.

PV requires a CONVEXITY ADJUSTMENT — the swap rate isn't a martingale under the forward measure → can't just discount expected forward swap rate.
Practical formula uses replication via swaptions (Hagan).

Use: curve steepening/flattening bets, hedging principal-protected notes. **Risk factors:** rate (multiple points on curve), vol (the convexity adjustment depends on the **entire swaption SABR vol surface**, not just one σ — change in skew matters).

E9. Asset Swap (ASW)

What: a PACKAGE — buy a bond and simultaneously enter an IRS that swaps the bond's fixed coupons for floating. Net result: synthetic floater earning $\text{LIBOR} + \text{ASW_spread}$.

ASW spread = fixed coupon - par swap rate (same tenor) (approx)

PV(package) = PV(bond) + PV(IRS at bond's coupon) = par by construction

Use: isolate the bond's credit-spread component from rate exposure. **Risk factors:** credit spread of bond, almost zero rate exposure post-swap.

E10. Year-on-Year (YoY) Inflation Swap

What: like ZCIS but pays YOY inflation each year, not a single compounded payment at maturity.

At each t_i : receive $N \cdot (\pi_{\{i\}} - K) \cdot \tau$ (annual inflation π_i)

Use: hedge year-by-year inflation exposure (vs lump-sum ZCIS for liability matching). **Risk factors:** breakeven curve, **inflation vol**, **inflation/rate correlation** (the YoY-vs-ZCIS convexity adjustment is $\rho \cdot \sigma_{\pi} \cdot \sigma_r \cdot \tau$ -flavoured — material at long tenors).

E11. Cap / Floor / Caplet

What: portfolio of interest-rate options. Cap = strip of caplets (each pays when the floating rate exceeds K); floor = puts; collar = cap + sold floor.

Caplet payoff at t_i : $\tau \cdot N \cdot \max(L_i - K, 0)$ paid at t_{i+1}
 Cap value = $\sum \text{Caplet}_i$
 Caplet pricing = Black formula on the FORWARD rate L_i^{fwd}

Use: hedge floating-rate borrowing (cap), hedge a bond portfolio against falling rates (floor). **Risk factors:** rate (delta), vol (vega) — caps are pure vol plays.

E12. Bermudan Swaption

What: swaption with MULTIPLE possible exercise dates (vs European = one). Holder can choose the optimal exercise.

Pricing requires LSMC (Longstaff-Schwartz) or lattice — no closed form
 Value > European swaption by the "early exercise premium"

Use: callable bond hedging (the embedded call is a Bermudan), structured products. **Risk factors:** rate, vol, **mean reversion** (which short-rate model you use matters).

E13. CDS Index (CDX / iTraxx)

What: standardised CDS contract on a basket of 100+ reference names (CDX NA IG = 125 names, US investment grade; iTraxx Europe = 125 names, European IG, etc.). Pre-defined coupons (100 bp for IG, 500 bp for HY).

Same math as single-name CDS (§15), but each defaulting name pays out a SLICE of the notional: payout per name = $(1/n) \cdot (1-R) \cdot N$.
 Remaining notional CONTINUES TO ACCRUE PREMIUM after each default.
 Index spread quoted in bp (e.g. CDX IG = 65 bp).

(Index $\neq$ first-to-default basket. FTD pays full notional on the FIRST default and terminates; the index pays incrementally and continues.)

Use: macro credit hedge, relative value vs single-name CDS, P&L from spread tightening. **Risk factors:** credit spread (CS01 of the whole index, often called "spread DV01"), correlation (for tranches), rate (small).

E14. Mortgage-Backed Security (MBS) — agency pass-through

What: bond backed by a pool of mortgages. Cashflows = scheduled principal + interest + UNCERTAIN prepayments (homeowners refinance when rates fall).

```
PV = Σ E[CFt | prepay model] · DF(t, spread)
      requires Monte Carlo over interest rate paths
      with a prepayment model (CPR/PSA)
```

Negative convexity — when rates fall, prepayments accelerate, you receive your money back at par EXACTLY when reinvestment yields drop.

Use: high-yield rate exposure with prepay risk (asset-mgr darling). **Risk factors:** - **rate** (effective DV01 — substantially lower than legal-maturity duration due to negative convexity) - **prepay risk** (CPR/PSA — separate scenario stress) - **OAS** (the spread that makes model PV = market PV after stripping the prepay option — NOT the same as vol) - **vol** (the rate-vol input INTO the prepay model — separately bumped to compute vega)

E15. Variance Swap

What: agree to exchange realised variance over $[0, T]$ for a fixed strike K^2 . Pure exposure to realised vol.

```
Payoff at T = N_var · ( σ²_realised - K² )      per VARIANCE notional N_var

Where:
  N_var      = variance notional ($ per unit of variance)
  N_vega     = vega notional ($ per vol point) = 2K · N_var

For small moves:  payoff ≈ N_vega · (σ_realised - K)      ← intuitive "$ per vol point"
```

Replication = a STATIC portfolio of out-of-the-money calls + puts (weighted as $1/K^2$) plus a continuous re-hedge of a delta-one position.

Use: vol arbitrage (realised vs implied), tail hedge (long variance = long crashes — non-linear payoff blows up under stress). **Risk factors:** vol (linear in variance, vs vanilla options which are concave in vol), correlation skew, jumps (gap risk).

E16. Equity Total Return Swap (Equity TRS)

What: same structure as bond TRS (§14) but reference is an equity (single stock or index). Receiver gets price changes + dividends; payer gets financing rate.

```
V = PV(equity total return) - funding_notional
```

Use: synthetic equity exposure (no balance sheet), tax-efficient long stocks. **Risk factors:** equity delta, equity vol (small), rate (financing leg), dividends.

E17. Commodity Forward / Swap

What: commodity equivalents of FX forward / IRS. Commodity forward locks in a future price; commodity swap pays floating index vs fixed.

```
Same math skeletons:
Forward:  V = N · (F_T - K) · DF(T)      where F_T = forward commodity price
Swap:    fixed leg vs floating commodity index, like an IRS but with comm index
```

Use: airlines hedge jet fuel (oil swap), corporates hedge raw materials. **Risk factors:** commodity price, basis (deliverable vs benchmark grade), seasonality, storage / convenience yield.

Universal Patterns to Memorise

Pattern A: Linear-in-strike

Applies to: **FX Forward (\$5), FRA (\$6), ZCIS (\$11), IRS (\$10 ≈ same), CDS (\$15).**

$$V = (\text{market_implied} - \text{contract}) \cdot \text{scale} \cdot \text{discount}$$

$$\text{Par strike} = \text{market_implied} \quad (\text{solves } V = 0)$$

Product	market-implied	scale	discount	par strike
FX Forward	$S \cdot \text{fx_factor}$	N_{for}	$DF(T)$	$S \cdot DF_{\text{for}}/DF_{\text{dom}}$
FRA	$\text{fwd} = (D_1/D_2 - 1)/\tau$	$N \cdot \tau$	$DF(T_2)$	fwd
ZCIS	$(1+b)^T$	N	$DF(T)$	b
IRS	$c_{\text{par}} = (1-D(T))/A$	$N \cdot A$	bundled	$(1-D(T))/A$
CDS	market_spread	$N \text{ (premium/protection)}$	$Q(t) \cdot DF(t)$	market_spread

Pattern B: basis × annuity

Applies to: **Sec Lending (\$3), Tenor Basis Swap (\$12), XCCY Basis Swap (\$13).**

$$V = \text{basis} \cdot \text{notional} \cdot \text{annuity_factor} \quad [\cdot \text{FX conversion if foreign}]$$

$$\text{annuity_factor} = \sum \tau \cdot DF(t)$$

Pure stream of small payments. Drill formula uses $b = \text{bp}/10000$.

Pattern C: Cashflow strip PV

Applies to: **Bond (\$7), Foreign Bond (\$8), Linker (\$9), Loan (\$4), Annuity Liability (\$18).**

$$V = \pm \sum \text{cf}_i \cdot \text{uplift}_i \cdot DF(t_i, \text{spread})$$

$$\text{uplift}_i = (1 + b)^{t_i} \quad \text{for inflation-linked}$$

$$= 1 \quad \text{for nominal}$$

$$\text{spread} = \text{issuer spread (0 for govt bonds, } > 0 \text{ for corp/credit-risky)}$$

$$\text{sign} = + \text{ for assets, } - \text{ for liabilities}$$

Risk Factor Quick Reference

Each "01" metric is a one-bp bump-and-revalue (rate factors) or scaled bump (FX, vol).

Factor	Affects	Risk metric	Bump-and-revalue
rate (bp)	every $DF(t)$	DV01	bump nominal curve +1 bp, reprice
breakeven (bp)	inflation-linked products (Linker, ZCIS, Annuity)	IE01	bump b +1 bp
credit spread (bp)	Bond, Loan, Foreign Bond, CDS	CS01	bump $spread_bp$ +1
FX (ret, e.g. 1%)	FXSpot, FXForward, ForeignBond, XCCYBasisSwap, FXOption	FX delta	scale FX legs by $(1 + 0.01)$
tenor basis (bp)	Tenor Basis Swap	tenor-01	bump that basis +1 bp
XCCY basis (bp)	XCCY Basis Swap	xccy-01	bump that basis +1 bp
vol (vol point)	FX Option, Swaption	vega	bump σ +1%

Per-instrument sign cheatsheet:

Long position in...	DV01 sign	IE01 sign	FX delta sign
Bond / Loan / Linker	– (rates $\uparrow \rightarrow P \downarrow$)	+ for Linker only	0 (unless foreign)
IRS — receive-fixed	+ (rates $\uparrow \rightarrow V \uparrow$)	0	0
IRS — pay-fixed	–	0	0
ZCIS — receive inflation	+ (rates $\uparrow \rightarrow V \uparrow$ via DF)	+	0
FX Spot / Forward (long foreign)	+ (via DF)	0	+
Annuity Liability	+ (V more negative when rates $\uparrow$)	–	0
Long-call FX/Swaption	small	0	+ (call) / – (put)

Annuity liability's signs are the OPPOSITE of a long bond's because the cashflows themselves are negative.

Convention Reminders (where people slip up)

1. **bp** → **decimal** is always $/ 1e4$. (15 bp = 0.0015.)
 2. **Inflation uses $(1+b)^t$ compounding**, not $1 + b \cdot t$ simple — inflation indices compound multiplicatively.
 3. **Year fraction τ** must match the contract's day-count (ACT/360, 30/360, ACT/ACT...) — drills use $\tau = 1.0$ annual but real desks rarely have $\tau = 1.0$ exactly.
 4. **Annuity liability NEGATIVE sign** — you OWE these. Forgetting reverses the entire ALM.
 5. **Linear-in-strike par rate**: set $V = 0$, divide out common factors → strike = market-implied. Works for FX Forward, FRA, ZCIS, IRS, CDS.
 6. **FX-sensitive products** (FX Spot, FX Forward, ForeignBond, XCCY, FX Option) need $S \cdot fx_factor()$; purely domestic products don't.
 7. **Forward in FRA** is the SIMPLE-compounded one: $(D_1/D_2 - 1)/\tau$ — used because FRA payoff $(L-K) \cdot \tau$ is itself simple. Continuous-compounded forward $-\ln(D_2/D_1)/(T_2-T_1)$ is for HJM / academic models.
 8. **CDS hazard $\lambda \neq$ spread s** : relationship is $s \approx \lambda \cdot (1-R)$. Use $\lambda = s/(1-R)$ to build the survival probability $Q(t) = \exp(-\lambda t)$.
 9. **Swap floating leg telescope**: $\Sigma \text{ payments}$ simplifies to $1 - DF(T)$ in single-curve world. In multi-curve world (LIBOR/OIS dual curve) you need both projection and discount curves — drill ignores this.
 10. **Bond spread** in this drill is **continuous-comp** ($\exp(-(z+s/1e4) \cdot t)$); production conventions also include Z-spread, OAS, ASW spread — see §7 caveats.
 11. **ZCIS quote convention**: K quoted as annualised CAGR (matches $(1+K)^T$). Some markets quote on continuous-comp basis — verify before trading.
 12. **Option vol** is per the BLACK model — lognormal in F (forward FX or forward swap rate), NOT in spot. Strike comparisons should always be vs the FORWARD, not the spot.
-

Appendix — Real-world ALM Caveats — what the drills gloss over

These are the things experienced desks know that the toy formulas hide. Worth a quick read when refreshing.

Multi-curve discounting (post-2008)

Drills use a single curve for discounting AND projection. Real desks:

- **Discount** at the **OIS** (or SOFR/€STR) curve — the "risk-free" funding rate
- **Project** floating rates on the LIBOR/SOFR-fixing curve (often different from OIS)
- Difference can be 30–50 bp under stress (e.g. 2008 LIBOR-OIS blowout)
- A real IRS therefore has two curves; the drill's single-curve $1 - DF(T)$ collapses this.

FRA settlement timing

Drill values FRA via the curve-implied forward, discounted at $DF(T_2)$. In real settlement:

- Cash payment occurs at T_1 (start of period), NOT T_2
- Payment = $N \cdot (L - K) \cdot \tau / (1 + L \cdot \tau)$ — discounted at the OBSERVED L , not the curve
- The two are equivalent under the T_2 -forward measure (martingale property)
- Practical importance: settlement discount matters for collateral calls

FRA vs Eurodollar/SOFR futures — convexity adjustment

A FRA and a STIR future on the same window are *almost* the same, but futures' daily mark-to-market creates a convexity bias of $\frac{1}{2}\sigma^2 T_1 T_2$. For 5y+ futures the adjustment is 5–20 bp — material when bootstrapping from futures vs FRAs.

XCCY swaps — principal exchange and MtM resets

Real XCCY swaps have **principal exchange at start and end** (omitted in drill). The "Mark-to-Market XCCY" variant **resets** the notional on every payment date based on the new spot — this dampens the FX P&L between fixings. Drill formula is the simplest possible version.

Linker — indexation lag

The CPI used at payment time is typically the index level from 3M earlier (UK index-linked gilts) or 8M earlier (US TIPS). Creates a small basis between contractual cashflows and the breakeven curve. Real IE01 has a lag-adjusted component.

Breakeven $\neq$ pure inflation expectation

Market breakeven $b \approx$ expected inflation + **inflation risk premium** + **liquidity premium** (Linkers are less liquid than nominals). For valuation it's b ; for forecasting actual inflation you should strip out the premia (typically 30–80 bp).

LDI / pension / insurer regulatory regimes

- **UK FRS102 / IFRS / IAS 19 / US ASC 715**: liabilities discounted on **AA-corporate curve**, not OIS
- $\rightarrow$ Liability DV01 vs swap DV01 differ by the **swap-vs-AA basis** (a real, persistent P&L source)
- LDI hedges nominal rate + inflation, but the AA-corporate basis is an UNHEDGED residual

- **Solvency II (EU insurers):** matching adjustment + volatility adjustment let you add a credit-spread bonus when matching cashflows; **SCR Interest Rate Risk** = parallel ± 64 bp scenarios on the risk-free curve (per EIOPA tables)
- **Basel III IRRBB (banks):** EVE and NII sensitivities under 6 prescribed scenarios (parallel $\pm$, short-rate $\pm$, steepener, flattener); HQLA classification affects which bonds count for LCR
- **Accounting:** AFS bonds $\rightarrow$ P&L volatility goes through OCI (insulates earnings); AC ("hold-to-maturity") $\rightarrow$ no MTM volatility but limits portfolio flexibility

Pension de-risking glide path

Most modern DB pension schemes operate a **funded-ratio-triggered glide path**: as the funded ratio (assets/liabilities) crosses thresholds, the hedge ratio steps up. Example: at 90% funded, hedge 60% of liability DV01; at 95% funded, step to 75%; at 100% funded, step to 95%. Mechanically forces de-risking when markets cooperate, leaves upside when they don't.

LPI caps and floors (UK pensions)

UK pensions often pay LPI (limited-price-indexation): inflation between 0% and 5%, capped at 5%. This makes the liability **non-linear in inflation** — ZCIS doesn't hedge the cap/floor convexity. Real LDI uses inflation caps + floors as well.

September 2022 gilt crisis — the liquidity risk of LDI itself

UK pension LDI hedges (long-duration rate swaps) required posting collateral when rates rose sharply. Forced selling of the assets being "hedged" cascaded into a gilt crisis. **Hedging strategies have their own liquidity profile** — match D_{mod} AND collateral capacity.

Multi-tenor basis swaps

Real desks have 3M-vs-6M, 1M-vs-3M, 1M-vs-6M basis swaps — each with its own quoted spread. The drill's "tenor basis swap" is the 3M-vs-6M flavour; production curves have a SURFACE of basis spreads.

TRS — who's exposed to what

The TRS receiver gets the bond's economic exposure (price + coupon) WITHOUT funding it on balance sheet. The TRS payer (typically a bank) provides synthetic financing and earns the funding spread. Common use: hedge funds use TRS to get bond exposure without using their balance sheet; banks use TRS to lay off counterparty exposure.

CDS — hazard rate vs market spread

Real CDS curves have a TERM STRUCTURE of spreads (1y, 3y, 5y, 10y), not one flat number. Bootstrap a hazard-rate curve from market spreads exactly like a discount curve. Drill assumes flat $\lambda \rightarrow$ 5y CDS only.

Nominal IRS + nominal bond not in the drill — but always paired with these

For a full hedge book you'd expect: - **Vanilla IRS** for nominal rate hedging (most-traded rates product on earth) - **Government bond** for risk-free cashflow projection - The drill's IRS (\$10) is exactly this — you have it. Just flagging that real ALM uses HEAPS of these alongside the others.

Real vs nominal rate decomposition (Fisher)

For inflation-linked products: $\text{nominal_rate} \approx \text{real_rate} + \text{breakeven}$. A 1 bp move in NOMINAL rates isn't the same as a 1 bp move in REAL rates — for a Linker, real-rate DV01 is what matters; for an annuity liability discounted on nominal, nominal DV01 matters. Don't conflate them.

That's the complete reference. The 6-block template per product means you can refresh ANY of these in 90 seconds: scan the formula, plug into the toy example, check the intuition, look up the closer.